

CineWave Entertainment

Case Lifecycle & System Architecture

1. Lifecycle Design Overview

The Movie Ticket Request case type utilizes Pega's Case Lifecycle Management (CLM) to represent the business journey. The case transitions through four sequential primary stages and concludes with automated resolution and notification.

Lifecycle Stage	Architectural Logic & Execution Flow
Stage 1: Submit Booking Request (Intake)	Collect Information step where customer inputs Movie Name, Show Date, Show Time, and Number of Tickets. Required field validations guarantee completeness.
Stage 2: Availability (Validation & Pricing)	Verifies show capacity using Seat Availability Status and Available Seats Count. Evaluates validation condition ($\text{Available Seats} \geq \text{Number of Tickets}$). Automatically computes $\text{Total Cost} = \text{Ticket Price} * \text{Number of Tickets}$.
Stage 3: Approval (Customer Decision)	Approve/Reject step assigned to Customer persona. Presents read-only summary review view. If approved, advances to Stage 4. If cancelled, transitions to Resolved-Cancelled.
Stage 4: Booking Execution (Fulfillment)	Automated step allocating physical Seat Numbers, updating Booking Confirmation Status to Confirmed, and issuing unique Ticket ID. Dynamically routes to respective work queue based on Show Type.
Resolution: Automated Correspondence	Upon reaching case resolution, an automated Correspondence rule triggers, extracting dynamic properties (.pyID, .MovieName, .ShowDate, .ShowTime, .NumberOfTickets, .SeatNumbers, .TotalCost) into a formatted confirmation email.

2. Flow & Decision Paths

- **Availability Validation Path:** If seats are insufficient, the case halts progression and notifies the user to select another screening.
- **Approval Branching:** Approval routes forward to Booking Execution; Rejection routes to immediate resolution with status Resolved-Cancelled.
- **Work Queue Routing Branching:** Premium shows route to PremiumShowQueue; standard shows route to StandardShowQueue.